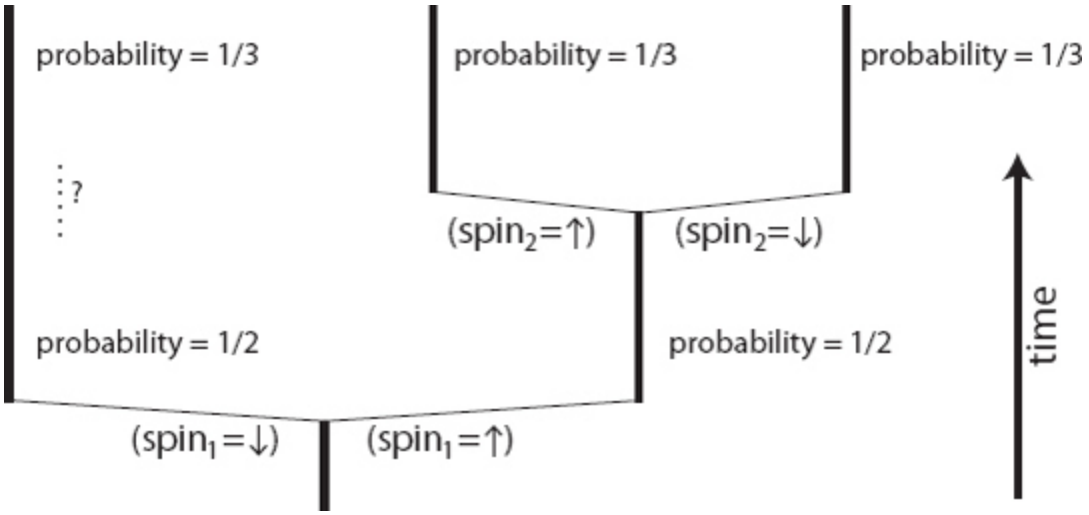


That's silly. If we followed that rule, the probability of the original spin-up branch would suddenly change when we did a measurement on the spin-down branch, going from 1/2 to 1/3. The probability of observing spin-up in our initial experiment shouldn't depend on whether someone on an entirely separate branch decides to do another experiment later on. So if we're going to assign credences in a sensible way, we'll have to be a little more sophisticated than simple branch counting.

Instead of simplistically saying "Assign equal probability to each branch," let's try something more limited in scope: "Assign equal probability to branches when they have equal amplitudes." For example, a single spin in a spin-right state can be written as an equal superposition of spin-up and spin-down.



This new rule says we should give 50 percent credence to being on either the spin-up or spin-down branches, were we to observe the spin along the vertical axis. That seems reasonable, as there is a symmetry between the